WESTMINSTER
International University in Tashkent

System Analysis & Design

Lecture 5

Agenda

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1. Last week recap
2. Extreme Programming (XP) introduction
3. XP values
4. XP principles
5. XP rules
6. Questions
7. Recommended reading

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Recap

What did we discuss last week?

Requirements Gathering

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1. What are the types of requirements?
2. Requirements gathering methods.
3. Steps in requirements gathering.

What is a requirement?

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Something that the product **must do** or a **quality** that the product **must have**.

1. **Functional requirements** – are things the product ***must do***.
   Examples: Ability to create a new blog post, edit existing posts, allow users to login, etc.

2. **Non-functional requirements** – are the ***qualities*** the product ***must have***.
   Examples: Prevailing color must be our corporate color, website should be able to process 100 concurrent users, interface must be easy to use, etc.

3. **Constraints** – are global issues that shape the requirements.
   Examples: Must be available in TAS-IX only, product must comply with ISO XXXX standard, legal constraints, etc.

NOTE: Collected/documented requirements must be unambiguous and non-vague.

Requirements Gathering Process

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1. Project Blastoff
2. Trawling for Knowledge
3. Prototyping and scenario modelling
4. Writing them down
5. Quality Gateway
6. Reusing requirements
7. Taking Stock of the specification
8. Post Mortem
9. Tailoring your process



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eXtreme Programming

Taking things to the extreme.

System Develop Methodologies

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Structural:

- Waterfall
- Structured Software Development Methodology (SSDM)

Iterative and Incremental:

- Spiral
- RAD
- eXtreme Programming
- Scrum
- Etc.

Prototyping:

- Prototyping

Taking things to their extremes…

That’s basically is the idea!

eXtreme Programming

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- First introduced in 1996 by Kent Beck at Chrysler's accounting system.

- Formulated and published as a software development methodology in 1999. Later, in 2004, Kent Beck reviewed the methodology and published the second edition of "eXtreme Programming explained" book. (We have several copies in our LRC.)

eXtreme Programming

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What is meant by “taking things to their extremes”?

Taking “**good practices**” to extreme.

For example:

- Code review improves code quality → make code review continues → Pair programming!
- Tests reduce errors, improve code → test everything → Cover all your code with unit tests, integration tests, etc.
- Etc.

XP: Values

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1. **Communication** – promote among the team members, also involve end users actively.
2. **Simplicity** – design the simplest solution. Code for today, not tomorrow. "You ain’t gonna need it" (YAGNI) approach.
3. **Feedback** – from unit tests, acceptance tests, customer.
4. **Courage** – to change your code or throw it away.
5. **Respect** – your peers. Don’t commit a change that breaks tests or compilation. Make sure everyone feels appreciated and heard.

XP: Practices

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Fine-scale feedback

- Pair programming
- Planning game
- Test-driven development
- Whole team

Continuous process

- Continuous integration
- Refactoring or design improvement
- Small releases

Shared understanding

- Coding standards
- Collective code ownership
- Simple design
- System metaphor

Programmer welfare

- Sustainable pace

XP: Rules

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Planning

- User stories are written.
- Release planning creates the release schedule.
- Make frequent small releases.
- The project is divided into iterations.
- Iteration planning starts each iteration.

Managing

- Give the team a dedicated open stand-up.
- Set a sustainable pace.
- A stand-up meeting starts each day.
- The Project Velocity is measured.
- Move people around.
- Fix XP when it breaks.

Design

- Simplicity.
- Choose a system metaphor.
- Use CRC cards for design sessions.
- Create spike solutions to reduce risk.
- No functionality is added early.
- Refactor whenever and wherever possible.

Coding

- The customer is always available.
- Code must be written to agreed standards.
- Code the unit test first.
- All production code is pair programmed.
- Only one pair integrates code at a time.
- Integrate often.
- Set up a dedicated integration computer.
- Use collective ownership.

Testing

- All code must have unit tests.
- All code must pass all unit tests before it can be released.
- When a bug is found tests are created.
- Acceptance tests are run often and the score is published.

XP: Discussion

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- What are the advantages of adopting XP?
- What are the disadvantages of adopting XP?



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Questions?

Recommended Reading

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- Beck, K. (2004). *Extreme programming eXplained*. 2nd ed. Reading, MA: Addison-Wesley.
- Extreme Programming: A gentle introduction, (2015). *The Rules of Extreme Programming*. [online] Available at: http://www.extremeprogramming.org/rules.html [Accessed 10 Nov. 2015].
- Georgia Tech (2015). *Software Development Processes*. [online] Udacity.com. Available at: https://www.udacity.com/course/software-development-process--ud805 [Accessed 10 Nov. 2015].
- Wikipedia, (2015). *Extreme programming*. [online] Available at: https://en.wikipedia.org/wiki/Extreme_programming [Accessed 10 Nov. 2015].

- *Google…*



The End